Written examination paper for SKY2100 – Cloud Security
Kristiania University College
Autumn 2022

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Exam format: Individual written home examination

Final report format: recommended LaTeX or Word format and font 12 with 1.0 spacing. The limit is 12 pages max, list of the bibliography can come in addition to this.

Grading scale: The Norwegian grading system uses the graded scale A - F, where A is the best grade, E is the lowest pass grade, and F is fail

Weighting: 100% of the overall grade

Support materials: All support materials are allowed

Plagiarism control: We expect your independent work. Please, use citations and quotations if there is material you want to include in the report.

Tasks format: answering task A is mandatory. Then, you can choose practical task B **OR** answer task C, in case your PC has some issues with virtualization. If you choose task C, please also include the screenshot of the error or system message (with time and data visible on the screenshot) showing that you cannot perform the virtualization task.

Learning outcomes as per course description:
Knowledge. The student...
- can define and explain how cloud services work
- can explain the terms Software as a Service, Platform as a Service and Security as a Service
- can define and explain how cloud solutions are secured and what differences and similarities this is in relation to internal servers
- can explain the difference between a Cloud Native application and a cloud migration that uses the Lift And Shift methodology
- can define and explain the role of a Web Application Firewall and how a WAF can be part of a security architecture
- can define and explain the development methodology DevSecOps and how this differs from traditional product development

Skills. The student...
- can use, configure and manage security in a cloud solution
- can install and configure a Web Application Firewall and use it to protect a cloud solution
- can use security and vulnerability analysis tools integrated with the cloud solution
- can use integrated logging in cloud solutions and understand how these logs can be integrated with a SIEM tool

General competence. The student can...

- reflect the architectural requirements and give and advice about security architecture and implementation of security solutions in cloud services
- describe and define the advantages and disadvantages of establishing or moving a project / service to the cloud from a security perspective

Exam Task

Task A (70% of the grade)

1. **Task A.1**: Write at least one-page essay explaining all the technologies you know in Cloud Computing and what is their primary purpose. Briefly, present also corresponding industry-grade cybersecurity (simple and efficient) protection mechanisms in modern cloud platforms that can reduce the number of cyber attacks and affiliated risks.
2. **Task A.2:** Imagine that you are hired by an international Software Development Company that is doing business in the industry related to the development of software modules, IoT nodes code snippets, Google- and MS Azure-based cloud platform integration. The company has established regional offices in Oslo, Beijing and Los Angeles, with all in-premises servers (only private cloud) located in these cities. The number of software developers currently involved in operations is about 100, and it is sufficient to use VPN and private cloud for internal projects and product development. The company is expected to hire another 4,000 software developers with even broader geographical coverage and establish ten more international offices next year. The private cloud solution needs to be spanned and integrated with major public cloud providers such as AWS, GCP and MS Azure to ensure 24/7 development and resilience in their business. **Task A.2.1.** Describe which cloud model and tiers you will use to accomplish this task. What type of cloud can you suggest for such a scenario? **Task A.2.2.** Present overview of the cyber threats and risks to the company's infrastructure (i) before the expansion, (ii) after the expansion. Explain what major disruptive events can occur because of such attacks and how they will impact the business. **Task A.2.3.** Name at least five different cyber security controls available after the expansion and how one can reduce the risks named in Task 3.2. Which controls will require the most financial investment and human resources to implement? **Task A.2.4.** Does the company need a private cloud after the expansion? If so, why?
3. **Task A.4:** Include a cost analysis screenshot from your "Microsoft Azure for Students" account "Home→Cost Management→Cost analysis" (with time and data visible on the screenshot).
4. **Task A.5:** Include a screenshot of your activity from GCP (with time and data visible on the screenshot): https://www.cloudskillsboost.google/profile/activity

Task B (30% of the grade)

1. Install Ubuntu / Debian Virtual Machine either in VMWare or VirtualBox.
2. In the command line, access information about a) the number of CPU/cores and bogomips per core, b) open network ports.
3. Install the Nginx server in your VM. Open two terminal windows and, in one, perform a simulation of the DDoS attack: "*ab -n 100000 -c 100 127.0.0.1*". In another, run the interactive "*htop*" command. What was the highest CPU load you managed to achieve? Include the screenshot in the report.
4. Install docker and run the same experiment in the docker container with Nginx. Include the screenshot in the report showing your docker installation, test results, and system load and compare if there are any differences with the results from (3).

5. Analyze the average number of requests/concurrency threads your VM machine can handle before it is offline (“DDoSed”) due to too many requests. Check if your VM can hold one million requests against the web server: “*ab -n 1000000 -c 1000 127.0.0.1*”

Task C (30% of the grade)

1. Imagine you need to design a private cloud computing platform for a company that operates with sensitive information – research of new and efficient medicine for cancer treatment. The company will need to process at least 100TBytes of data and work across several offices in Norway distributed in multiple geographical regions. Describe in detail the infrastructure you will build for this task, components, software platform, and cybersecurity protection mechanisms that might be relevant. Do you need any mechanisms such as IAM, encryption or DDoS protection? Why?